07. Excel Files with OpenPyXL

Detailed beginner handbook - English and Arabic

Technologies: OpenPyXL

What you will learn

Create and read a synthetic Excel workbook.

OpenPyXL lets Python create, read, and validate .xlsx sheets and cells without opening Excel.

الهدف: بناء مثال صغير وفهمه خطوة بخطوة.

شرح بسيط: تعلم الفكرة أولا، ثم جرب المثال الصغير ببيانات اصطناعية فقط.

How to study

Study one lesson at a time. Type examples yourself, predict the result, run the code, then change one thing. Keep a notebook of new words, errors, root causes, and fixes.

Course map

Setup and mental model; ten core concepts; a guided mini-project; the real dashboard architecture; debugging; exercises and answers.

Lesson 1 - Setup and mental model

Prerequisites

Windows 10 or 11, VS Code, PowerShell, and permission to install learning dependencies. Use only synthetic data in tutorials and screenshots.

Setup sequence

1. Open PowerShell.
2. Go to learning-examples/07-excel-openpyxl.
3. Read README.md.
4. Run pip install -r requirements.txt; python workbook.py.
5. Read the complete error if it fails.
6. Change one safe value and rerun.

الخطوات: افتح مجلد المثال، شغل الأمر، غير قيمة واحدة، ولاحظ النتيجة.

Mental model

Treat OpenPyXL as one layer in a system. Identify the input, the transformation or rule, the output, and possible failures. Understanding that flow is more valuable than memorising syntax.

Checkpoint

Explain: What problem does this technology solve? What is its input? What output should it produce? What can fail?

Lesson 2 - Workbook, worksheet, row, column, and cell concepts

What is it?

An .xlsx workbook contains worksheets; each worksheet contains rows and columns; a cell has a coordinate, value, and optional style or formula.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Excel Files with OpenPyXL, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
from openpyxl import Workbook
book=Workbook(); sheet=book.active
print(sheet.title, sheet['A1'].value)
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Workbook, worksheet, row, column, and cell concepts.

هذا الدرس يشرح: Workbook, worksheet, row, column, and cell concepts. اكتب المثال بنفسك، حدد المدخلات والعملية والنتيجة، ثم غير قيمة واحدة لتفهم السلوك.

Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 3 - Creating workbooks and appending typed rows

What is it?

Workbook creates a file, active selects the default sheet, title renames it, append adds one row, and save writes the ZIP-based .xlsx package.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Excel Files with OpenPyXL, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
sheet.title='Summary'
sheet.append(['Month','Total'])
sheet.append(['Jan',12])
book.save('demo.xlsx')
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Creating workbooks and appending typed rows.

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Quick check

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Lesson 4 - Opening files safely in normal, read-only, and data-only modes

What is it?

load_workbook opens an existing file. read_only streams large sheets; data_only reads cached formula results instead of formula text when Excel has saved those results.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Excel Files with OpenPyXL, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
from openpyxl import load_workbook
book=load_workbook('demo.xlsx', read_only=True, data_only=True)
print(book.sheetnames)
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Opening files safely in normal, read-only, and data-only modes.

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Quick check

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Lesson 5 - Reading headers and validating expected sheet structure

What is it?

Read the first row as headers, normalise harmless whitespace, reject missing required columns, and report unexpected or duplicate headers before analysing rows.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Excel Files with OpenPyXL, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
headers=[cell.value for cell in sheet[1]]
required={'Month','Total'}
missing=required-set(headers)
if missing: raise ValueError(f'Missing: {missing}')
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Reading headers and validating expected sheet structure.

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Quick check

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Lesson 6 - Dates, numbers, formulas, blank cells, and type surprises

What is it?

Excel dates may be datetime objects or numbers with formats. Blank cells are None. Numeric identifiers can lose leading zeroes, so define expected column semantics.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Excel Files with OpenPyXL, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
from datetime import datetime
sheet.append([datetime(2026,1,1), 12])
for row in sheet.iter_rows(values_only=True): print(row)
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Dates, numbers, formulas, blank cells, and type surprises.

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Quick check

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Lesson 7 - Styles, number formats, widths, filters, and frozen panes

What is it?

Styles control fonts, fills, borders, alignment, and number formats. Column widths, filters, and frozen headers improve a human-facing export but do not change data meaning.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Excel Files with OpenPyXL, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
from openpyxl.styles import Font
sheet['A1'].font=Font(bold=True)
sheet.freeze_panes='A2'
sheet.auto_filter.ref=sheet.dimensions
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Styles, number formats, widths, filters, and frozen panes.

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Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 8 - Iterating efficiently and avoiding unnecessary memory use

What is it?

iter_rows(values_only=True) avoids creating unnecessary cell objects. Read only required sheets and columns, especially for large uploads.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Excel Files with OpenPyXL, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
for month,total in sheet.iter_rows(min_row=2, values_only=True):
    print(month,total)
# values_only avoids unnecessary cell objects.
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Iterating efficiently and avoiding unnecessary memory use.

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Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 9 - Writing formulas versus reading cached formula values

What is it?

Writing '=SUM(B2:B5)' stores a formula. OpenPyXL does not calculate it; Excel or another calculation engine must generate the cached result.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Excel Files with OpenPyXL, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
sheet['B3']='=SUM(B2:B2)'
book.save('formula.xlsx')
# OpenPyXL stores this formula but does not calculate it.
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Writing formulas versus reading cached formula values.

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Quick check

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Lesson 10 - In-memory workbooks with BytesIO and upload validation

What is it?

BytesIO keeps an uploaded workbook in memory. Validate ZIP structure and size before parsing, and return clear errors for missing or malformed sheets.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Excel Files with OpenPyXL, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
from io import BytesIO
stream=BytesIO(upload_bytes)
book=load_workbook(stream, read_only=True)
# Validate archive size before parsing.
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates In-memory workbooks with BytesIO and upload validation.

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Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 11 - Corrupt files, unsafe archives, privacy, and export testing

What is it?

Treat spreadsheets as untrusted input. Reject macros when unsupported, limit archive expansion, exclude sensitive fields from exports, and reopen generated files in tests.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Excel Files with OpenPyXL, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
try:
    book=load_workbook('upload.xlsx',read_only=True)
except Exception as exc:
    raise ValueError('Invalid .xlsx workbook') from exc
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Corrupt files, unsafe archives, privacy, and export testing.

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Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 7 - Guided mini-project

Project goal

Create and read a synthetic Excel workbook.

Starting example

```
from openpyxl import Workbook
book=Workbook(); sheet=book.active
sheet.append(['Month','Total']); sheet.append(['Jan',12])
book.save('demo.xlsx')
```

Read the code

Find imported tools or page elements. Identify the synthetic input. Find the function, event, route, or transformation that changes it. Finally, locate where the result is displayed, returned, saved, or packaged.

Build sequence

1. Run the untouched example.
2. Record the result.
3. Rename one unclear value.
4. Add a synthetic record.
5. Validate empty input.
6. Validate incorrect input.
7. Add a helpful error.
8. Rerun every case.
9. Compare with exercise-solution.
10. Explain every line.

Definition of done

Normal, empty, and invalid cases behave deliberately; no private data is used; and you can explain every line without reading the guide.

Lesson 8 - The real dashboard

Architecture connection

The application combines OpenPyXL with the rest of the stack. The frontend gathers filters and files; the local API validates requests; the data layer transforms workbook rows; charts present aggregates; export tools create files; and the desktop layer packages the application for Windows.

Trace the upload feature

The user selects a workbook. The frontend sends it to the local API. The backend validates the file and sheets. The data layer creates safe tables. The API returns metadata. React updates state. Plotly displays results. Identify exactly where this guide's technology participates.

Privacy and quality

Do not place narrative, contact, or identifying fields in tutorials, logs, screenshots, tests, or exports. Use generated identifiers and synthetic totals. Validate at each boundary and collect only fields required for the calculation.

Architecture exercise

Draw five boxes: UI, API, Data Processing, Export, Desktop Packaging. Add arrows and label the data. Circle the box related to this guide and add two validation checks.

Lesson 9 - Debugging

Reliable debugging loop

1. Reproduce with the smallest input.
2. Read the first meaningful error.
3. Find the exact file and line.
4. Inspect value and type.
5. Compare expected and actual results.
6. Change one thing.
7. Rerun the same case.
8. Add a regression test.

Common beginner mistakes

- Wrong folder or command.
- Missing dependency or inactive environment.
- Misspelled file, variable, field, route, or import.
- Assuming input is always present.
- Mixing setup, processing, and display in one block.
- Hiding an exception rather than understanding it.
- Using real sensitive data in a test.

أخطاء شائعة: تشغيل الأمر من مجلد خاطئ، نسيان تثبيت التبعيات، أو تغيير أشياء كثيرة بمرة واحدة.

Error notebook

Record: command, expected result, actual error, root cause, smallest fix, and test added. This turns errors into reusable knowledge.

Lesson 10 - Exercises and answers

Exercises

1. Explain the technology in three sentences.
2. Recreate the example without copying.
3. Add one normal synthetic record.
4. Handle empty input.
5. Handle invalid input.
6. Improve unclear names.
7. Add or describe one test.
8. Locate the technology in the dashboard.
9. Record one error and root cause.
10. Add one mini-project feature.

تمرين: أضف شهرا أو قيمة آمنة جديدة، ثم اشرح ما حدث بكلماتك.

Answer guide

A strong solution is observable and explainable: the documented command works; output changes for the new record; empty and invalid cases have deliberate results; names communicate purpose; a test states an expected result; and the architecture explanation identifies the correct boundary.

Next step	Next: 08. Data Analysis